MIL-S-20427
27 November 1951

MILITARY SPECIFICATION

SHELL, STEEL, MANUFACTURED FROM FORGINGS, FOR MEDIUM CLAIBER RECOILLESS RIFLES

This specification was approved by the Departments of the Army, the Navy, and Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-30-8
19 October 1948

This specification consists of this cover sheet and Specification 50-30-8, dated 19 October 1948, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-30-8
19 October 1948

SHELL, STEEL, MANUFACTURED FROM FORGINGS, FOR MEDIUM CALIBER RECOILLESS RIFLES

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawings pertaining to the shell under contract and all drawings and specifications listed thereon shall form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of the invitation for bids.

B. CLASSES AND TYPES.

B-1. This specification covers the manufacture of shell from forgings and includes all point-fuzed and base-fuzed High Explosive and Chemical Shell, for 75mm and 105mm Recoilless Rifles.

C. MATERIAL AND WORKMANSHIP.

C-1. All material and parts shall be in accordance with applicable specifications and drawings.

C-2. The shell shall be machined from forgings manufactured in accordance with Specification No. 57-104-2, "Forgings for Common Steel Shell and Shrapnel."

C-3. The machining and treatment of the shell in each lot shall be uniform.

C-4. The interior of all shell shall be free from burrs, scale, fins, deep scars or pits, and sharp edges. The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance.

C-5. All shell shall be sound and free from cracks and porosity.

C-5a. The liner of HEAT Shell shall be smooth and free from scores, pits, and other surface imperfections, which affect the serviceability of the liner. Slight evidence of the successive draws (draw rings) on the surfaces of the liner is not an objectionable imperfection.

C-6. Base cover welds.- Base cover welds shall be sound, and the welding operations shall be so controlled that there will be adequate fusion between the base of the shell and the base cover.

C-7. Cleaning prior to Painting.- Projectile surfaces to be painted shall be cleaned and prepared in accordance with Grade I, Specification JAN-C-490. These cleaned surfaces shall not be recontaminated prior to painting. After painting these surfaces shall meet the various performance requirements of Specification JAN-C-490.

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C-8. Coating. The exterior coating and the cavity coating specified on the drawing for the shell shall be applied in such manner as to produce a tightly adherent coating of uniform thickness over the entire specified areas. The shell shall be stenciled or marked as prescribed on the drawing. All paint and marking shall be dry to the touch before shell are packed for shipment.

C-9. Banding.—

C-9a. Band Seat. Prior to banding, the band seats shall be clean and free from oil, grease, dirt, rust and other foreign material. Shot or sand blasting is not permitted for cleaning band seats after knurling.

C-9b. Rotating Band.—

C-9b(1). The rotating band blank shall be in the form of a solid ring, annealed if necessary to meet the requirements of Specification No. 50-27-1, except that in the case of hot banding at not less than 800°F, no prior annealing is necessary. The band blank shall be carefully pressed into the band seat so as to completely fill it and fit tight. (See Paragraph F-5).

C-9b(2). The band shall be applied in such a manner as not to distort the walls of the shell. Distortion shall not be considered to have taken place if the band seat and the shell body immediately adjacent to the band are within drawing tolerances specified.

C-9b(3). The bands shall be applied cold.

C-10. Protection of Rotating Band. The machined and engraved rotating bands shall not be nicked, burred, or otherwise damaged in subsequent handling and shipping.

D. GENERAL REQUIREMENTS.

D-1. Forgings.—

D-1a. A forging lot shall contain no more than one heat of steel. Small lots (such as groups of forgings which have become separated from previously delivered lots, hospitalized forgings, etc.,) may be grouped into miscellaneous forging lots at the discretion of the machiner. Such lots shall be identified by a miscellaneous lot code number. The size of the miscellaneous forging lot shall not exceed 100 Tons.

D-1b. During machining operations the steel heat code number or miscellaneous lot code number shall be marked on each forging following any machining operation which removes it, so that the lot may be identified until it has passed final physical tests.

D-2. Shell requiring a nosing-in operation may be nosed-in either hot or cold. The contractor shall establish a method satisfactory to the contracting officer for maintaining control of the interior ogive contour of the shell.

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E. DETAIL REQUIREMENTS.

E-1. The physical properties for the shell covered by this specification shall be as specified on the applicable drawings, and may be obtained by heat treatment if necessary.

E-2. The material of the rotating band shall show in test specimens taken from bands removed from the shell, the following minimum physical properties:-

T. S. - p.s.i.	%.Elon - 1"	Bend Test Degrees
32,000	15	180°

E-3. Steel for Forgings. The material shall be of such quality and chemical composition so as to meet requirements of WDSS-6 steel of Specification No. 57-104-1.

F. METHODS OF SAMPLING, INSPECTION AND TEST.

F-1. Inspection. Inspection shall be as specified in the Specification No. 50-0-1, General Specification for Ammunition Except Small Arms Ammunition, and Specification No. QQ-M-151, Metals: General Specification for Inspection of.

F-2. Lots and Lot Numbers:-

F-2a. Pilot Lot. A pilot lot of 100 shell shall be furnished as provided in Specification No. 50-0-1. Twenty-five shell from the pilot lot shall be inert loaded and ballistically tested at 112% of the maximum rated pressure of the weapon concerned. Results of the ballistic test will be carefully scrutinized by the Ordnance Department to determine the quality of the shell.

F-2b. Shell Lots. Upon completion of all physical tests, the lots shall be regrouped for future identity into shell lots to which will be assigned serial shell lot number. The maximum size of a lot shall be 20,000 shell.

F-2c. Serial numbers for shell lots shall begin with the number "1" and shall continue serially for each contractor in such a way that there will never be a duplication of numbers used by the same contractor on the same caliber of shell. This shell lot serial number shall appear stamped on the finished shell as prescribed in Specification No. 50-0-1.

F-3. Tension Test.—

F-3a. Two shell shall be selected by the inspector from each forging lot, for tension tests. Two tension test specimens, conforming to Type I, Specification No. QQ-M-151 or (if the wall thickness of the shell is not sufficient for type 1) to the largest size of type 4 specimen obtainable, shall be taken longitudinally from opposite sides of the wall of each of the sample shell. The test specimen shall be taken in such position that the gage length will include the section of the wall at the location of the rotating band, and in such manner that the test coupon is from within the contours of the finished shell.

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F-3b. The yield strength shall be determined by either the offset method or the extension-under-the-load method, as prescribed in Specification No. QQ-M-151. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length). In addition the yield point shall be determined by the divider method described in Specification No. QQ-M-151. The results shall be recorded but shall not be basis for acceptance of the shell.

F-3c. Retests. In case of failure to meet physical requirements, a retest of specimens from double the number of shell shall be made upon request of the contractor.

F-4. Soundness.—

F-4a. Each shell shall be inspected for cracks and porosity after finish machining and before application of the base cover. All shell which show cracks or porosity shall be rejected. Doubtful shell shall be rejected except that at the option of the contractor they may be hydrostatically tested as provided in F-4b.

F-4b. If the contractor elects to test doubtful shell hydrostatically, he shall provide hydraulic equipment capable of testing shell to an internal pressure of 25,000 pounds per square inch. Each doubtful shell shall be subjected to hydraulic pressure sufficient to stress the metal of the body to the prescribed yield strength of the material. The hydraulic pressure to be used shall be determined by the following formula:

$$P = \frac{2St}{D}$$

where P = pressure in pounds per sq. in.
S = fiber stress in pounds per sq in.
t = maximum allowable wall thickness at forward bourrelet
D = bourrelet diameter in inches

F-5. Rotating band tests.—

F-5a. Detailed Inspection.—

F-5a(1). Method of Inspection. One band per 2,000 shell from each banding machine (or where production is less than 2,000 per working shift, one band from each banding machine from each shift) shall be removed for examination of the seating.

F-5a(2). Character of Seating. Each band removed in accordance with paragraph F-5a(1) shall be examined for character of seating and to determine whether corners of the band seat have been properly filled.

F-6. Brazing Test.—

F-6a. For chemical shell that require the adapter to be silver brazed to the body, all requirements of the drawing shall be complied with. In order to ascertain that the silver solder has penetrated to the depth prescribed on the drawing for 360°, the nose portion of one shell from each unit of 2,000 produced shall be sectioned into quarters. In addition, all shell of each unit of 2,000

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which fail to pass the air pressure test shall be similarly sectioned, provided that not more than 10 shell of the unit fail the air pressure test. If more than 10 shell of the unit of 2,000 fail the air pressure test, then the entire 2,000 shell represented by the failures shall be rejected.

F-6b. If any surface exposed by the sectioning process described in Paragraph F-6a shows less than the prescribed penetration of silver solder, a retest of twice the number originally sectioned shall be made if requested by the contractor. If any surface of the nose portions sectioned in the retest fails to show the minimum penetration of silver solder, then the entire 2,000 shell represented by the tests shall be rejected.

F-6c. Shell rejected in accordance with paragraphs F-6a and F-6b above may be reworked by the contractor and shall be accepted if they fulfill all requirements of the drawing and specification, including paragraphs F-6a and F-6b above.

G. PACKING AND MARKING.

G-1. Packing.—

G-1a. Shell shall be packed on pallets as specified on Drawings 76-18-5, 75-18-5A, 75-18-7, 76-0-48.

G-1b. False ogive shell, shall be packed, as directed by the Office, Chief of Ordnance, in commercial type containers so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

G-1c. Data Cards. Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 lb. ream weight, shall be prepared for each lot of accepted projectiles to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind _____ Lot No. _____

Quantity in Lot _____

Drawing No.* _____ Spec. No.* _____

Manufactured by _____ Date _____

Contract No. _____ Order No. _____

Packed _____

Remarks _____

Certified to by _____ Inspector
(Print here name of inspector, District, Arsenal, or Loading Plant, and Sub-District where applicable)

* Include revision and amendments

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G-2. Marking. The marking for shipment shall be in accordance with Specification No. 100-2, "Standard Specification for Marking Shipments by Contractors".

H. NOTES.

H-1. The necessity for firmly seating the rotating band so that the band is in contact with the shell at all points cannot be over emphasized. Bands must be fully and uniformly annealed prior to pressing and pressure must be adequate and uniformly applied in order to obtain satisfactory results.

H-2. This specification replaces Ordnance Department Tentative Specification AXS-1496.

H-3. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

NOTE: Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

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